

ASSIGNMENT 3 – Group 2_106

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I. Data analysis

First we need to analyze the whole data set to see if all the regions evolve in the same way and if we can see stationarity

```
> global <- ts(Cement[3:8], start = 1996, frequency = 12)
> autoplot(global)
```

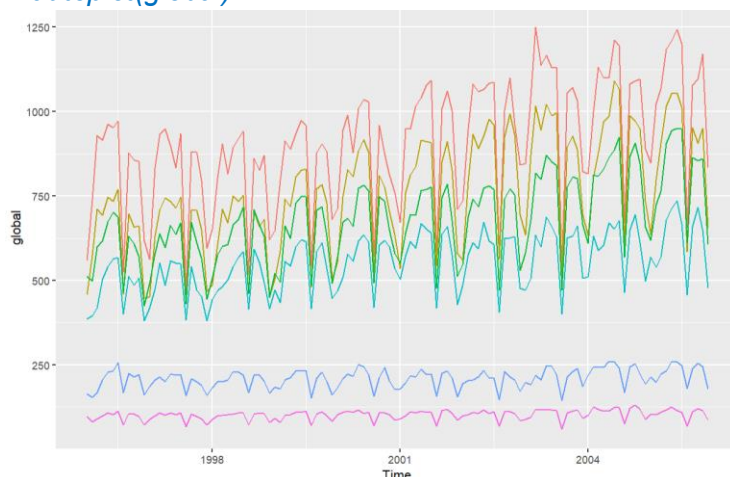


Figure 1: autoplot(global)

All the regions seem to have seasonality, which is probably due to the fact that the need for electricity vary during day/night or summer/winter, but we see different level of variability for each region which could be explained by the fact that all regions don't have the same weather, therefore different electricity needs throughout the year. The variance of each region seems to increase over time which let us think that there is a non-stationarity in the variance, so we will use a multiplicative model.

A slight trend seems to appear for NOV, NES, CEN and SUD probably linked to the fact that we know human tend to consume more and more electricity as time is passing. However, we observe 2 regions, SIC and SAR, that evolve differently than the others regarding a potential trend. We decomposed SIC and SAR and discovered that they clearly have a trend. There is definitely a non-stationarity in the mean.

```
> autoplot(decompose(SAR, type = c("multiplicative")))
```

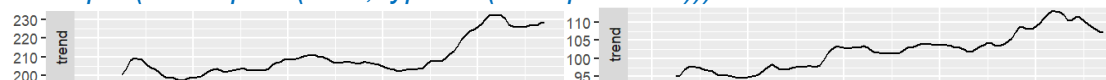


Figure 2: Decomposition of SIC (left) and SAR (right)

Based on those information, we decided to forecast each region separately and then sum up the results. It will give the company a more accurate estimate taking in account the different trends, speed and variability of needs of each region. As we will do a cross-validation procedure for our forecast, we will separate our data in 2 sets for each region.

```
> NOVtraining <- ts(NOV, start = 1996, end = c(2003,02), frequency = 12)
> NOVtest <- window(NOV, start = c(2003,03), end = c(2005, 12))
```

II. Forecasting

Which model to choose:

Moving average: NO as seasonality is very strong and will influence too much.

Simple exponential regression: NO as there is non-stationarity

Linear exponential smoothing: NO because consider only trend

Winter exponential smoothing: YES because consider trend + seasonality

Simple/ Multiple linear regression: NO because no clear driver available

ARIMA

A. Winter exponential smoothing

Alpha α : high variability, no shift in the local mean → lower alpha (0.1-0.3)

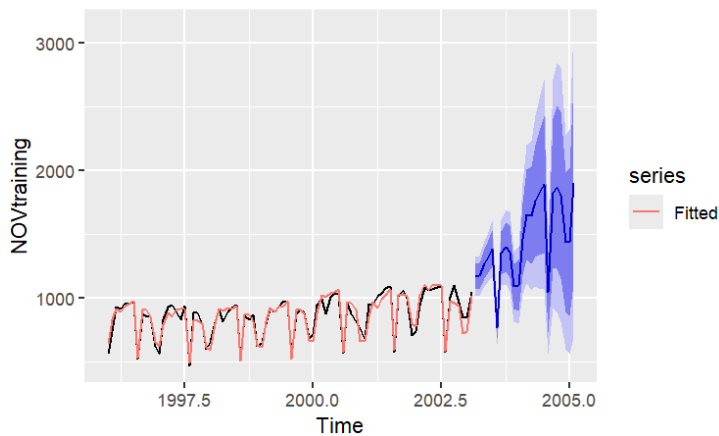
Beta β (trend): long term trend $\rightarrow$ low beta (0.1 -0.3)

Gamma γ (seasonality): constant long-term seasonality $\rightarrow$ low gamma (0.1 -0.3)

Test 1: our choices

```
fsNOVtraining <- hw(NOVtraining, alpha=0.1, beta=0.1, gamma=0.1, seasonal = "multiplicative")
autoplot(fsNOVtraining) + autolayer(fitted(fsNOVtraining), series = "Fitted")
```

Forecasts from Holt-Winters' multiplicative method



```
> accuracy(fsNOVtraining)
```

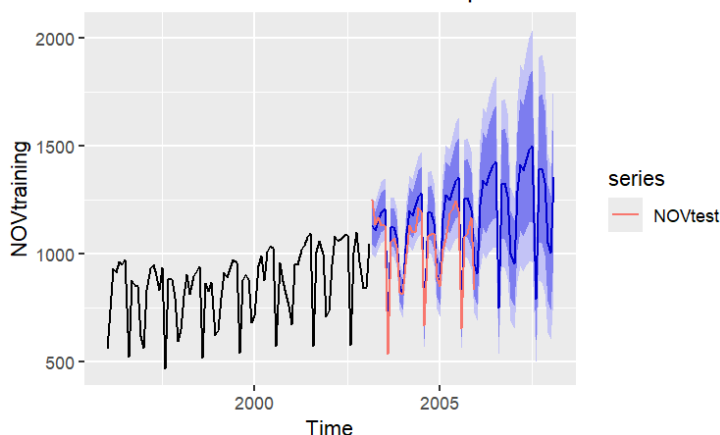
	ME	RMSE
MAE		
Training set:	1.413981	37.08364
	MPE	MASE
ACF1		
	0.0868649	0.3113816

Regarding the Mean Error and Mean absolute error numbers, it's not that bad considering the unit of measure.

Test 2: automatic choice of parameters

```
fsautoNOVtraining <- hw(NOVtraining, seasonal = "multiplicative", h = 60)
autoplot(forecast(fsautoNOVtraining)) + autolayer(NOVtest)
```

Forecasts from Holt-Winters' multiplicative method



```
> accuracy(fsautoNOVtraining)
```

	ME	RMSE
MAE		
Training set:	8.149458	34.91719
	MPE	MASE
ACF1		
	0.843857	0.6706132
	0.1377011	

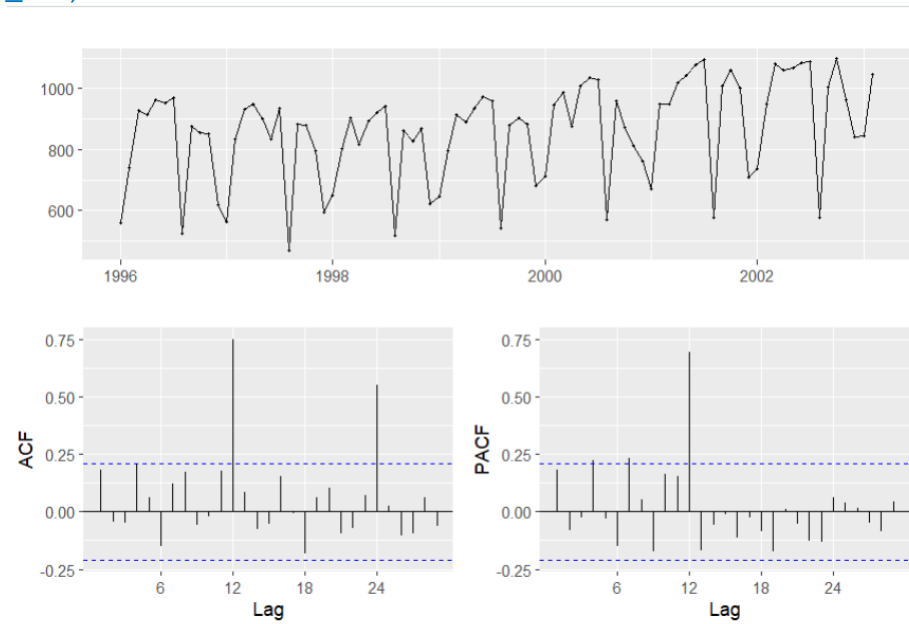
```
> Str(fsautoNOVtraining)
```

```
:List of 4
..$ value : num 1842
..$ par   : num [1:16] 0.10649 0.01262 0.00013 808.51687 -4.58359 ...
```

The system decreased beta and gamma and seems to have a better fit than our Test 1.

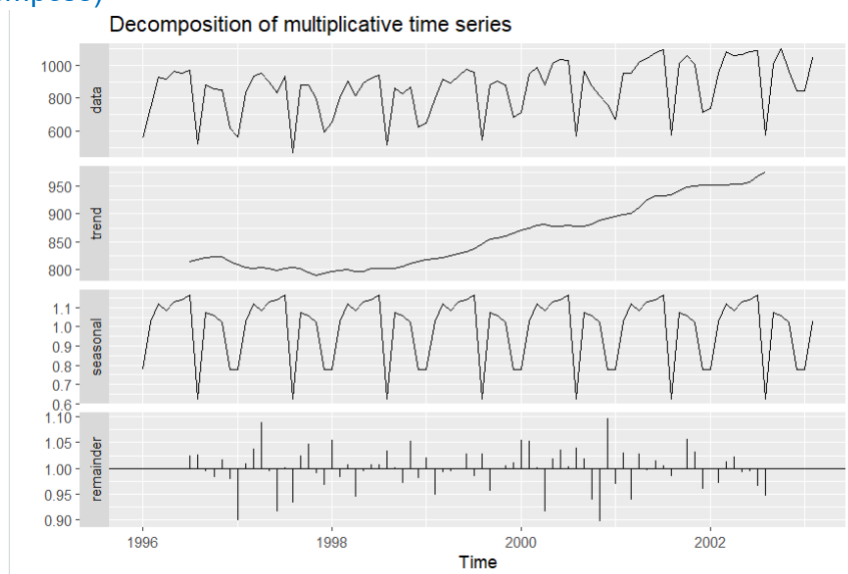
B. Arima model (individual series)

```
> ggtsdisplay(train_nov)
```



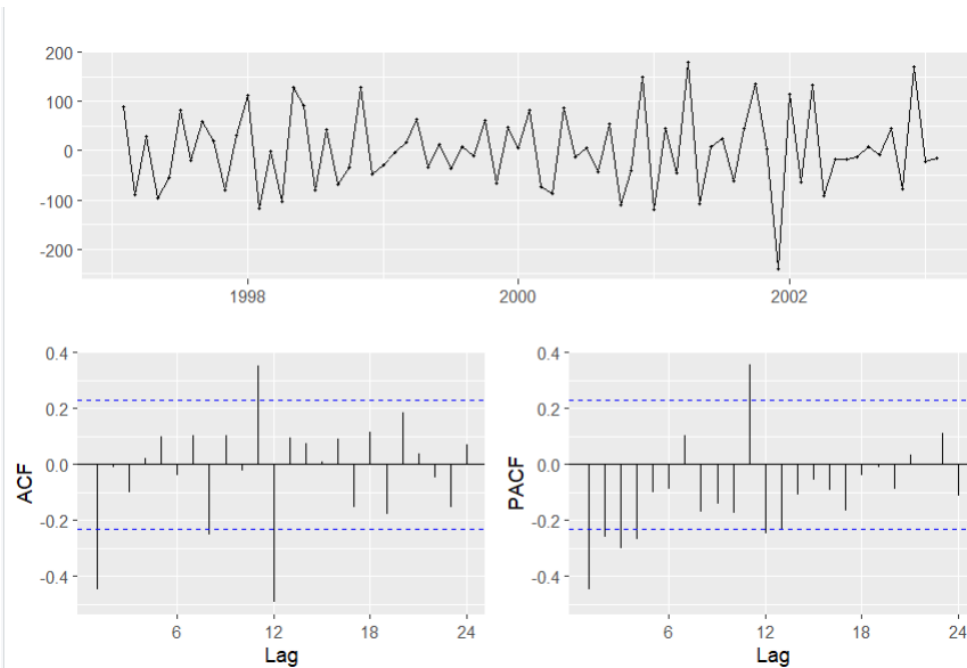
We need to check the seasonality and order of the series, since it's quite difficult to find from this, we do the decomposition.

```
> nov_decompose <- decompose(train_nov, type = "multiplicative")  
> autoplot(nov_decompose)
```



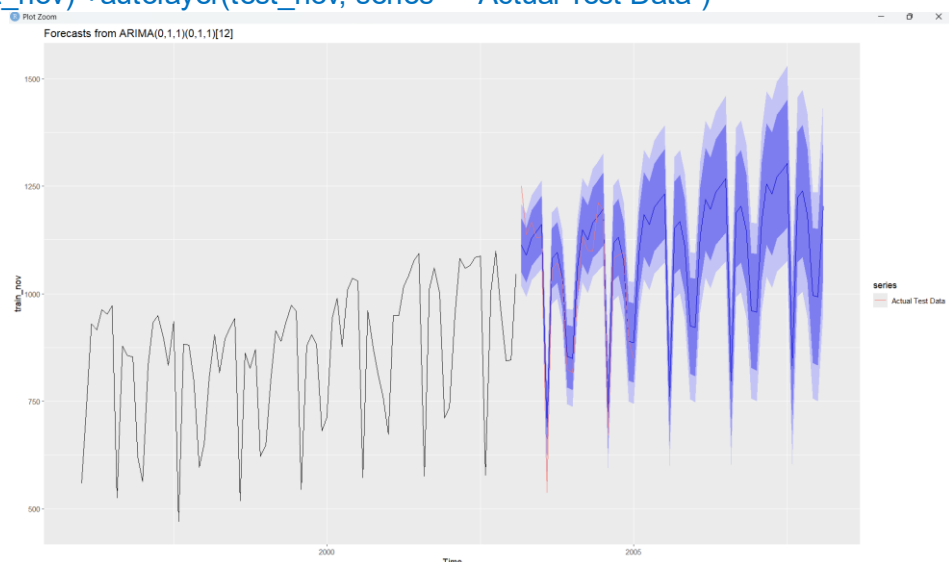
We can observe both trend and seasonality in the time series. To apply an ARIMA model, the series must first be made stationary. This is achieved by removing the trend and seasonal patterns through differencing.

```
> diffT_nov <- diff(train_nov, lag = 1, differences = 1)  
> diffT_nov2 <- diff(diffT_nov, lag = 12, differences = 1)  
> ggtsdisplay(diffT_nov2)
```



After applying first-order and seasonal differencing, the series appears to be approximately stationary. The ACF plot shows a significant spike at lag 1, followed by a rapid decline, which suggests a non-seasonal moving average component of order 1 (MA(1)). In addition, the ACF displays a notable spike at lag 12, indicating the presence of a seasonal moving average component of order 1. The PACF tails off gradually, which further supports the selection of a moving average structure rather than an autoregressive one. Therefore, a suitable model for the series is $ARIMA(0,1,1)(0,1,1)[12]$. We can see an exponential decrease at the PACF which indicate it's a moving average series and the peaks in the ACF shows that it is of order 1 and seasonality 1.

```
>forecast_nov <- forecast(nov_model, h = 60)
autoplot(forecast_nov) +autolayer(test_nov, series = "Actual Test Data")
```

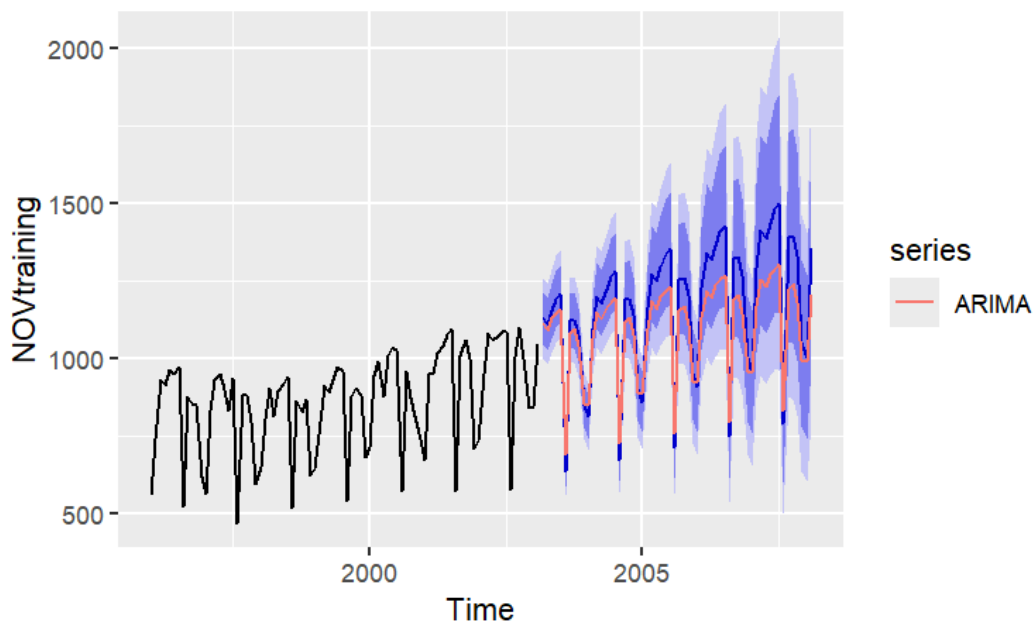


C. Comparison

Graphic view:

```
autoplot(fsautoNOVtraining) + autolayer(forecast_nov $mean, series = "ARIMA")
```

Forecasts from Holt-Winters' multiplicative method



Accuracy Holt Winters model:

```
> accuracy(fsautoNOVtraining, NOVtest)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	8.149458	42.91789	34.91719	0.843857	4.225875	0.6706132	0.1377011
Test set	-68.185062	92.78635	77.81265	-6.719875	7.515058	1.4944556	0.2998809
	Theil's U						
Training set	NA						
Test set	0.3429677						

Accuracy Manual ARIMA model Accuracy

```
> accuracy(forecast_nov, test_nov)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	5.384564	44.29445	33.09283	0.4668341	3.946956	0.6355749	0.06690089	NA
Test set	-16.631989	54.51372	40.58637	-2.5363392	4.534754	0.7794944	0.25704188	0.1543853

Conclusion:

We selected **ARIMA model** because it shows the best performance on the test set, with significantly lower errors (RMSE, MAE, MAPE) than the Holt-Winters model, indicating a better ability to generalize.

III. Forecast for 2006-2007

NOV. ARIMA (0,1,1) (0,1,1) [12]

>write.table(forecast_nov, file = "forecast_nov.csv", quote = F, sep = ";", dec = ".")

Date	Point.Fore	Lo.80	Hi.80	Lo.95	Hi.95
01-01-2006	921	808	1034	747	1095
01-02-2006	1133	1018	1248	957	1308
01-03-2006	1219	1100	1338	1037	1401
01-04-2006	1197	1076	1317	1012	1381
01-05-2006	1237	1114	1359	1050	1424
01-06-2006	1252	1128	1376	1062	1442
01-07-2006	1268	1142	1393	1075	1460
01-08-2006	797	669	924	602	992
01-09-2006	1189	1060	1318	991	1386
01-10-2006	1203	1072	1334	1003	1403
01-11-2006	1146	1014	1279	944	1349
01-12-2006	961	827	1095	756	1166
01-01-2007	957	821	1092	749	1164
01-02-2007	1168	1031	1306	959	1378
01-03-2007	1255	1114	1396	1039	1471
01-04-2007	1232	1089	1375	1014	1451
01-05-2007	1272	1128	1417	1051	1494
01-06-2007	1288	1141	1434	1063	1512
01-07-2007	1303	1155	1452	1076	1530
01-08-2007	833	682	983	603	1062
01-09-2007	1225	1073	1377	992	1457
01-10-2007	1239	1085	1392	1004	1474
01-11-2007	1182	1026	1337	944	1420
01-12-2007	996	839	1154	756	1237
Total	27472	24212	30732	22487	32457

Forecast CEN. ARIMA (0,1,1) (0,1,1) [12]

Date	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
01/01/06	634	560	707	522	746
01/02/06	730	656	804	617	843
01/03/06	789	710	867	668	909
01/04/06	771	691	851	649	894
01/05/06	844	763	925	720	968
01/06/06	852	770	934	726	978
01/07/06	850	767	933	722	978
01/08/06	566	482	651	437	695
01/09/06	823	737	908	692	954
01/10/06	838	751	924	705	970
01/11/06	779	691	867	645	913
01/12/06	610	521	698	474	745
01/01/07	650	561	740	513	787
01/02/07	747	656	838	608	886
01/03/07	805	710	901	659	951
01/04/07	788	691	885	640	936
01/05/07	861	763	959	711	1011
01/06/07	869	769	968	717	1020
01/07/07	867	766	967	713	1020
01/08/07	583	481	685	427	738
01/09/07	839	736	942	682	997
01/10/07	854	750	959	695	1014
01/11/07	796	690	901	634	957
01/12/07	626	520	733	463	789
Total	18370	16193	20547	15040	21700

Forecast SUD. ARIMA (0,1,1) (0,1,1) [12]

Date	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
01/01/06	480	365	595	304	656
01/02/06	529	412	647	350	709
01/03/06	596	471	720	406	786
01/04/06	579	452	707	385	774
01/05/06	649	519	779	451	848
01/06/06	627	494	760	424	830
01/07/06	615	480	751	408	823
01/08/06	414	276	552	203	625
01/09/06	619	478	760	404	834
01/10/06	625	482	769	406	844
01/11/06	595	449	741	372	818
01/12/06	467	318	615	240	694
01/01/07	481	330	631	250	711
01/02/07	530	377	683	296	764
01/03/07	597	436	757	352	841
01/04/07	580	417	743	331	830
01/05/07	650	484	817	396	905
01/06/07	628	458	797	369	887
01/07/07	616	444	789	352	880
01/08/07	415	239	590	146	683
01/09/07	620	441	798	347	893
01/10/07	626	445	807	349	903
01/11/07	596	412	780	314	877
01/12/07	467	280	654	182	753
Total	13601	9960	17242	8033	19169

Forecast SIC. ARIMA (1,0,0) (0,1, 1) [12]

Date	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
01/01/06	190	171	210	161	220
01/02/06	196	177	215	166	226
01/03/06	209	188	229	178	239
01/04/06	212	192	232	181	243
01/05/06	236	216	256	205	267
01/06/06	223	202	243	192	254
01/07/06	220	200	240	189	251
01/08/06	154	134	174	123	185
01/09/06	223	202	243	191	254
01/10/06	224	203	244	193	255
01/11/06	206	185	226	174	237
01/12/06	166	146	186	135	197
01/01/07	190	170	211	159	222
01/02/07	196	176	216	165	227
01/03/07	209	187	230	176	241
01/04/07	212	191	233	180	245
01/05/07	236	215	257	204	269
01/06/07	223	201	244	190	255
01/07/07	220	199	241	187	252
01/08/07	154	133	175	122	187
01/09/07	223	201	244	190	255
01/10/07	224	202	245	191	256
01/11/07	206	184	227	173	238
01/12/07	166	145	187	133	198
Totale	4915	4420	5410	4159	5671

Forecast NES. – ARIMA (0,1,1) (0,1,1) [12]

Date	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
01/01/06	819	690	948	621	1016
01/02/06	1043	911	1174	842	1243
01/03/06	1150	1010	1290	936	1364
01/04/06	1130	987	1273	911	1349
01/05/06	1186	1040	1333	962	1410
01/06/06	1215	1065	1365	986	1444
01/07/06	1203	1051	1356	970	1437
01/08/06	827	671	982	589	1065
01/09/06	1155	997	1314	913	1397
01/10/06	1199	1038	1361	953	1446
01/11/06	1134	970	1298	883	1385
01/12/06	918	751	1084	662	1173
01/01/07	876	707	1046	617	1136
01/02/07	1100	928	1272	836	1363
01/03/07	1207	1026	1388	931	1484
01/04/07	1187	1003	1372	905	1469
01/05/07	1244	1056	1432	956	1531
01/06/07	1272	1081	1464	980	1565
01/07/07	1261	1066	1455	963	1559
01/08/07	884	686	1082	581	1187
01/09/07	1212	1011	1414	904	1521
01/10/07	1257	1052	1461	944	1570
01/11/07	1191	984	1399	874	1509
01/12/07	975	764	1186	652	1298
Total	26646	22543	30749	20371	32921

Forecast SAR. ARIMA (1,0,0)(0,1,2)[12] with drift

Date	Point.Forec	Lo.80	Hi.80	Lo.95	Hi.95
01-01-2006	99	92	107	88	111
01-02-2006	105	98	113	94	117
01-03-2006	116	108	124	104	128
01-04-2006	114	106	122	102	126
01-05-2006	122	114	130	110	134
01-06-2006	115	107	122	103	127
01-07-2006	117	109	125	105	129
01-08-2006	77	69	85	65	89
01-09-2006	119	111	127	107	131
01-10-2006	121	113	129	109	133
01-11-2006	113	105	121	101	125
01-12-2006	93	86	101	81	105
01-01-2007	101	93	109	89	113
01-02-2007	107	99	115	95	119
01-03-2007	118	109	126	105	130
01-04-2007	116	107	124	103	128
01-05-2007	123	115	132	111	136
01-06-2007	116	108	124	103	129
01-07-2007	118	110	127	106	131
01-08-2007	78	70	86	65	91
01-09-2007	120	112	129	108	133
01-10-2007	122	114	130	109	135
01-11-2007	114	106	122	101	127
01-12-2007	95	87	103	82	108
TOTAL	2640	2448	2832	2346	2934

TOTAL ITALY:

Our forecast considering all the regions is **93 643**.

Date ▼	NOV ▼	NES ▼	CEN ▼	SUD ▼	SIC ▼	SAR ▼
Jan 2006	921	819	634	480	190	99
Feb 2006	1133	1043	730	529	196	105
Mar 2006	1219	1150	789	596	209	116
Apr 2006	1197	1130	771	579	212	114
May 2006	1237	1186	844	649	236	122
Jun 2006	1252	1215	852	627	223	115
Jul 2006	1268	1203	850	615	220	117
Aug 2006	797	827	566	414	154	77
Sep 2006	1189	1155	823	619	223	119
Oct 2006	1203	1199	838	625	224	121
Nov 2006	1146	1134	779	595	206	113
Dec 2006	961	918	610	467	166	93
Jan 2007	957	876	650	481	190	101
Feb 2007	1168	1100	747	530	196	107
Mar 2007	1255	1207	805	597	209	118
Apr 2007	1232	1187	788	580	212	116
May 2007	1272	1244	861	650	236	123
Jun 2007	1288	1272	869	628	223	116
Jul 2007	1303	1261	867	616	220	118
Aug 2007	833	884	583	415	154	78
Sep 2007	1225	1212	839	620	223	120
Oct 2007	1239	1257	854	626	224	122
Nov 2007	1182	1191	796	596	206	114
Dec 2007	996	975	626	467	166	95
TOTAL	27472	26646	18370	13601	4915	2639